

Feynman Rules for $\lambda\phi^4$ from the operator formalism.

$$\bullet \langle 0 | T(\phi_1 \phi_2 \phi_3 \phi_4) | 0 \rangle = \phi_1 \phi_2 \phi_3 \phi_4 + \phi_1 \phi_2 \phi_3 \phi_4 + \phi_1 \phi_2 \phi_3 \phi_4$$

$$= D_F(x_1, x_2) D_F(x_3, x_4) + D_F(x_1, x_3) D_F(x_2, x_4) + D_F(x_1, x_4) D_F(x_2, x_3)$$

$$1 \bullet \quad \bullet 3 \Rightarrow \text{diagrams: } \text{two vertical lines} + \text{two horizontal lines} + \text{crossed lines}$$

$$2 \bullet \quad \bullet 4 \quad S_F = \langle f | S | i \rangle, \quad S = U_I(t, \infty)$$

* We are propagating the state $|i\rangle$ from $-\infty$ to ∞ and then computing the probability of ending in the state $|f\rangle$

$$\bullet \text{ Recall: } \langle \Omega | \dots | \Omega \rangle = \lim_{T \rightarrow \infty} \frac{U_I(t_0, -T) | \Omega \rangle}{e^{iE_0(T+t_0)} \langle \Omega | \Omega \rangle} = \lim_{T \rightarrow \infty} \frac{\langle 0 | T \{ \phi_I(x_1) \dots \phi_I(x_n) e^{-i \int dt H_I} \} | 0 \rangle}{\langle 0 | T \{ \exp[-i \int_T^T dt H_I] \} | 0 \rangle}$$

* Take ϕ^4 theory with:

$$\langle \Omega | T \{ \phi_x \phi_y \} | \Omega \rangle = \lim_{T \rightarrow \infty} \frac{\langle 0 | T \{ \phi_x \phi_y e^{-i \int dt H_I} \} | 0 \rangle}{1}$$

$$\int d^4x \mathcal{H}_I = \int \int d^3x \frac{\lambda}{4!} \phi^4 dt$$

$$\therefore -i \int dt \int d^3x \frac{\lambda}{4!} \phi^4 = -\frac{\lambda}{4!} \int d^4x \phi^4$$

Let us look at the Numerator:

$$\langle 0 | T \{ \phi_x \phi_y \exp(-i \lambda/4! \int d^4z \phi^4) \} | 0 \rangle = \langle 0 | \phi_x \phi_y | 0 \rangle + \frac{-i\lambda}{4!} \int d^4z \langle 0 | T \{ \phi_x \phi_y \phi_z^4 \} | 0 \rangle$$

$$\bullet \text{---} \bullet + \left(\frac{-i\lambda}{4!} \right) \alpha \text{---} \bullet + \left(\frac{-i\lambda}{4!} \right) \beta \quad \bigcirc$$

Here $n=3$: there are three ways to do this contraction for ϕ_z^4 .

$\beta = 12$:

In full we can write this as

$$\langle 0 | T \{ \phi_x \phi_y \bar{c} \} | 0 \rangle = 3 \left(\frac{-i\lambda}{4!} \right) D_F(x-y) \int d^4 z D_F(z-x) P_F(z-x) + 12 \left(\frac{-i\lambda}{4!} \right) \int d^4 z D_F(x-z) D_F(y-z) D_F(z-z)$$

Look at $\mathcal{O}(\lambda^3)$:

$$\langle 0 | T \{ \phi_x \phi_y \frac{1}{3!} \left(\frac{-i\lambda}{4!} \right)^3 \int d^4 z d^4 w d^4 u \phi_u^4 \phi_w^4 \phi_z^4 \} | 0 \rangle$$

$$\phi_x \phi_y (\phi_u \phi_u)^2 (\phi_z \phi_z)^2 (\phi_w \phi_w)^2$$

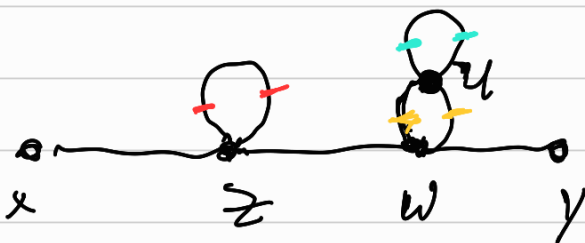
$$\phi_x \phi_y \phi_u \phi_u \phi_u \phi_u \phi_z \phi_z \phi_z \phi_z \phi_w \phi_w \phi_w \phi_w$$



+ ... (vac... Bubbles)

$$\phi_x \phi_z \phi_y \phi_w \phi_z \phi_w (\phi_u \phi_u)^2 \phi_z \phi_z \phi_u \phi_u$$

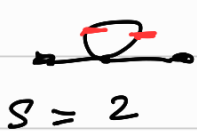
symmetry factor:



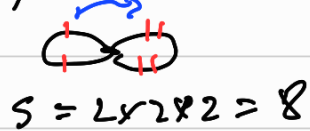
$$\frac{3! \times (4!)^3}{8!}$$

(Symmetry factor) $\equiv \frac{\text{denominator}}{\# \text{ of contractions}} = \frac{P! (4!)^3}{\# \text{ of contr.}}$

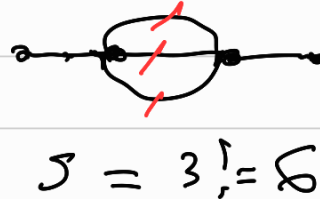
• Symmetry factor of the above: $2 \times 2 \times 2 = 8$



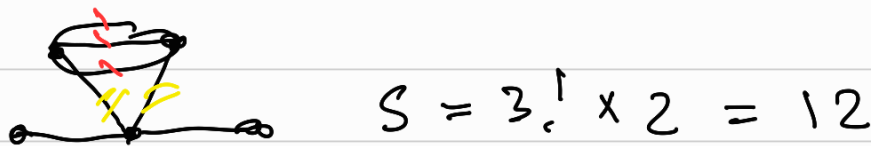
$$S = 2$$



$$S = 2 \times 2 \times 2 = 8$$



$$S = 3! = 6$$



$$S = 3! \times 2 = 12$$

* X-space Feynmann Rules for $\lambda \phi^4$:

internal line: $(-i\lambda) \int d^4 z_i$

$$D_F(x-y) = \int \frac{d^4 p}{(2\pi)^4} \left(\frac{-i}{p^2 + m^2 - i\epsilon} \right) e^{ip(x-y)}$$

momentum space propagator:

$$D_F(p) = -i / (p^2 + m^2 - i\epsilon)$$

* $V_1 \equiv$ , $V_2 \equiv$ 

etc....

$\rightarrow$ Then terms in the expansion of $\langle \Omega | T \phi_{x_1} \dots \phi_{x_n} | \Omega \rangle$ are:

$$\lim_T \langle 0 | T \{ \phi_{x_1} \phi_{x_2} \dots \phi_{x_n} e^{-i \int dt H_I(t)} \} | 0 \rangle$$

$$= (\text{Connected}) \sum_{\{n_i\}} \prod_i \frac{1}{n_i!} (V_i)^{n_i}$$

$$= (\text{Connected}) \left(\sum_{n_1} \frac{1}{n_1!} (V_1)^{n_1} \right) \left(\sum_{n_2} \frac{1}{n_2!} (V_2)^{n_2} \right) \times \dots$$

$$= (\text{Connected}) \prod_i \left(\sum_{n_i} \frac{1}{n_i!} (V_i)^{n_i} \right)$$

$$= () \prod_i e^{V_i}$$

$$= (\text{Connected}) e^{\sum_i V_i}$$

$$\lim_{\epsilon \rightarrow 0} \frac{1}{2\pi(1-i\epsilon)} \langle 0 | T \{ e^{-i \int dt H(i)} \} | 0 \rangle = e^{\sum_i V_i}$$

and so we conclude:

$$\langle \Omega | T \{ \phi(x) \phi(y) \} | \Omega \rangle = \sum (\text{connected diagrams})$$

ex:  $\propto \lambda \phi^4$

$$(-i\lambda)^4 \int d^4 z_1 \dots d^4 z_4 D_F(k - z_1) D_F(z_1 - z_2) D_F(z_1 - z_3) D_F(z_2 - z_4) D_F(z_3 - z_4)$$

$$|\text{Symm}| = 2 \times 2 \times 3! = 24$$